

MIKE (DEYUAN) HE

EDUCATION



PRINCETON UNIVERSITY, Princeton, NJ

2022 – 2027

Ph.D. in Computer Science

Thesis Advisor: Prof. Aarti Gupta

Research Focuses: Programming Languages; Formal Methods; Distributed and Agentic Systems

W University of Washington, Seattle, WA

2018 – 2022

B.S. in Computer Science

Advisors: Prof. Zachary Tatlock & Dr. Steven Lyubomirsky

Selected Honor: CRA Outstanding Undergraduate Researcher Award, Honorable Mention (2022)



SELECTED PUBLICATIONS

- **SPECY: Learning Specifications for Distributed Systems from Event Traces.** Mike He, Ankush Desai, Aishwarya Jagarapu, Doug Terry, Sharad Malik, Aarti Gupta. *OOPSLA*, 2026.
- **ALGOVERI: An Aligned Benchmark for Verified Code Generation on Classical Algorithms.** Haoyu Zhao*, Ziran Yang*, Jiawei Li*, Mike He*, Zenan Li*, Chi Jin, Venugopal Veeravalli, Aarti Gupta, Sanjeev Arora. *ICML*, 2026.
- **Goedel-Code-Prover: Hierarchical Proof Search for Open State-of-the-Art Code Verification.** Zenan Li*, Ziran Yang*, Deyuan He, Haoyu Zhao, Andrew Zhao, Shange Tang, Kaiyu Yang, Aarti Gupta, Zhendong Su, Chi Jin. *COLM*, 2026.
- **Theory-Level Autoformalization: From Isolated Statements to Unified Formal Knowledge Bases.** Marcus J. Min, Mike He, Zhaoyu Li, Zixuan Yi, Sharad Malik, Aarti Gupta, Xujie Si, Osbert Bastani. *ICML*, 2026.
- **Dynamic Tensor Rematerialization.** Marisa Kirisame*, Steven Lyubomirsky*, Altan Haan*, Jennifer Brennan, Mike He, Jared Roesch, Tianqi Chen, Zachary Tatlock. *ICLR*, 2021.

</> EXPERIENCE

Amazon Web Services, Santa Clara, CA & Boston, MA

Summer Intern 2024, 2025 & 2026

Applied Scientist Intern, Agentic Automated Reasoning

- Developed PINFER that automatically learns specifications for distributed systems (2024);
- Contributed to PVERIFIER that formally verifies correctness of distributed protocols (2025);
- Contributed to Strata, enabling concurrency reasoning in Strata&Lean (2026).

Taichi Graphics, Remote and Beijing, China

June. 2022 – Sep. 2022

Compiler R&D Intern

- Implemented tensor/vector types for Taichi language.

Intel, Remote and Hillsboro, OR

Mar. 2022 – June 2022

Formal Verification Research Intern at Strategic CAD Lab

- Developed PYROPE, a framework for formally verifying Python programs leveraging Dafny.



SKILLS

- **Languages:** Natural Language Prompts, Python, C/C++, C#, Rust, L^ΛEXN (open to others)
- **Applied PL:** Equality Saturation, SMT and Theorem Proving
- **Systems:** Compilers and Distributed Systems
- **Fun Fact:** I am more experienced in classical music than coding 🎵. I have been playing the violin for >20 yrs, with 6 years of experience in a philharmonic orchestra.